

13

Section and Mid-Point Formula

13.1 Introduction :

For any two known (given) points in a co-ordinate (Cartesian) plane, the knowledge of co-ordinate geometry may be used to find :

- the distance between the given points,
- the co-ordinates of a point which divides the line joining the given points in a given ratio,
- the co-ordinates of the mid-point of the line segment joining the two given points,
- equation of the straight line through the given points,
- equation of the perpendicular bisector of the line segment obtained on joining the given two points, etc.

13.2 The Section Formula :

To find the co-ordinates of a point which divides the line segment joining two given points in a given ratio.

(If a point P lies in a line segment joining the points A and B, then P divides AB in the ratio AP : PB).

Let AB be a line joining the points $A = (x_1, y_1)$ and $B = (x_2, y_2)$ and point P divides the line segment AB in the ratio $m_1 : m_2$.

$$\text{i.e.} \quad \frac{AP}{PB} = \frac{m_1}{m_2}$$

Required to find : The co-ordinates of point P.

Let $P = (x, y)$

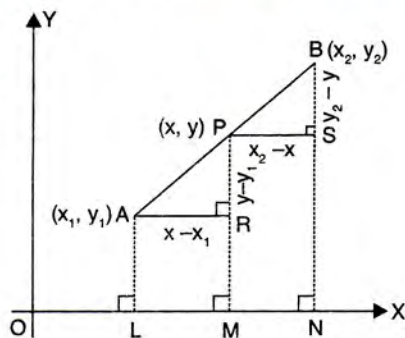
Draw AL, PM and BN perpendiculars on the x-axis. Thus, AL, PM and BN are parallel lines. It is clear from the figure that :

$$AR = LM = OM - OL = x - x_1;$$

$$PR = PM - RM = PM - AL = y - y_1;$$

$$PS = MN = ON - OM = x_2 - x$$

$$\text{and, } BS = BN - SN = BN - PM = y_2 - y$$



Since, ΔAPR and ΔPBS are similar.

$$\therefore \frac{AR}{PS} = \frac{PR}{BS} = \frac{AP}{PB} \quad [\text{Corresponding sides of similar } \Delta \text{ s are in proportion}]$$

$$\begin{aligned} \frac{AR}{PS} = \frac{AP}{PB} &\Rightarrow \frac{x - x_1}{x_2 - x} = \frac{m_1}{m_2} \\ &\Rightarrow m_2x - m_2x_1 = m_1x_2 - m_1x \quad [\text{By cross multiplication}] \\ &\Rightarrow m_1x + m_2x = m_1x_2 + m_2x_1 \\ &\Rightarrow x(m_1 + m_2) = m_1x_2 + m_2x_1 \\ \therefore x &= \frac{m_1x_2 + m_2x_1}{m_1 + m_2} \end{aligned}$$

Since,

$$\frac{PR}{BS} = \frac{AP}{PB} \Rightarrow \frac{y - y_1}{y_2 - y} = \frac{m_1}{m_2} \Rightarrow y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$$

$$\therefore \text{Co-ordinates of } P = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$$

- 1** Find the co-ordinates of point P which divides the join of A (4, -5) and B (6, 3) in the ratio 2 : 5.

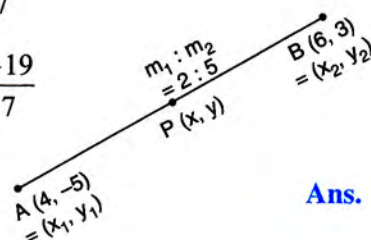
Solution :

Let the co-ordinates of P be (x, y)

$$\therefore x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2} = \frac{2 \times 6 + 5 \times 4}{2 + 5} = \frac{32}{7}$$

$$\text{and, } y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2} = \frac{2 \times 3 + 5 \times -5}{2 + 5} = \frac{-19}{7}$$

$$\therefore P = \left(\frac{32}{7}, \frac{-19}{7} \right)$$



Ans.

Conversely, to find the ratio in which the line joining the two points is divided by a given point.

- 2** Find the ratio in which the point (5, 4) divides the line joining points (2, 1) and (7, 6).

Solution :

Let the required ratio be $m_1 : m_2$

Take (2, 1) = (x_1, y_1) ; (7, 6) = (x_2, y_2) and (5, 4) = (x, y)

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2} \Rightarrow 5 = \frac{m_1 \times 7 + m_2 \times 2}{m_1 + m_2}$$

$$\Rightarrow 5m_1 + 5m_2 = 7m_1 + 2m_2$$

$$\Rightarrow 2m_1 = 3m_2$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{3}{2}$$

∴ The required ratio is 3 : 2.

Ans.

Alternative method :

In order to find the ratio in which the join of two given points is divided by a third point, take $m_1 : m_2 = k : 1$.

By doing so, two unknowns m_1 and m_2 are reduced to one unknown i.e. k and the section formula becomes :

$$x = \frac{kx_2 + x_1}{k + 1} \text{ and } y = \frac{ky_2 + y_1}{k + 1}$$

$$m_1 : m_2$$

$$= \frac{m_1}{m_2} : \frac{m_2}{m_2}$$

$$= k : 1$$

$$\therefore k = \frac{m_1}{m_2}$$

Let the required ratio be $k : 1 (= m_1 : m_2)$.

$$\therefore x = \frac{kx_2 + x_1}{k + 1} \Rightarrow 5 = \frac{k \times 7 + 2}{k + 1}$$

$$\Rightarrow 5k + 5 = 7k + 2$$

$$\Rightarrow 2k = 3$$

$$\Rightarrow k = \frac{3}{2}$$

∴ The required ratio = $k : 1 = \frac{3}{2} : 1 = 3 : 2$

Ans.

- 3** In what ratio is the line joining the points (4, 2) and (3, -5) divided by the x-axis ? Also, find the co-ordinates of the point of intersection.

Solution :

Let the required ratio be $k : 1$ and the point on the x-axis be $(x, 0)$.

Since, $y = \frac{ky_2 + y_1}{k + 1}$

[Taking (4, 2) = (x_1, y_1) and (3, -5) = (x_2, y_2)]

$$\Rightarrow 0 = \frac{k \times -5 + 2}{k + 1}$$

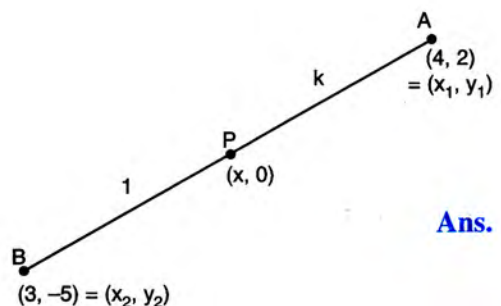
$$\Rightarrow 0 = -5k + 2$$

$$\Rightarrow k = \frac{2}{5}$$

$$\Rightarrow m_1 : m_2 = 2 : 5$$

Now, $x = \frac{2 \times 3 + 5 \times 4}{2 + 5}$

$$= \frac{26}{7}$$



Ans.

∴ The ratio = 2 : 5 and the required point of intersection = $\left(\frac{26}{7}, 0\right)$

Ans.

- 4** Calculate the ratio in which the line joining the points (4, 6) and (-5, -4) is divided by the line $y = 3$. Also, find the co-ordinates of the point of intersection.

Solution :

The co-ordinates of every point on the line $y = 3$ will be of the type $(x, 3)$.

$$\text{Now, } y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \quad [\text{Taking : } (x, 3) = (x, y), (4, 6) = (x_1, y_1) \text{ and } (-5, -4) = (x_2, y_2)]$$

$$\Rightarrow 3 = \frac{m_1 \times -4 + m_2 \times 6}{m_1 + m_2}$$

$$\Rightarrow 3m_1 + 3m_2 = -4m_1 + 6m_2 \Rightarrow 7m_1 = 3m_2 \Rightarrow \frac{m_1}{m_2} = \frac{3}{7}$$

$\therefore$ The required ratio is 3 : 7

Ans.

$$\text{Now, } x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \Rightarrow x = \frac{3 \times -5 + 7 \times 4}{3 + 7} = \frac{13}{10}$$

$\therefore$ The required point of intersection = $\left(\frac{13}{10}, 3\right)$

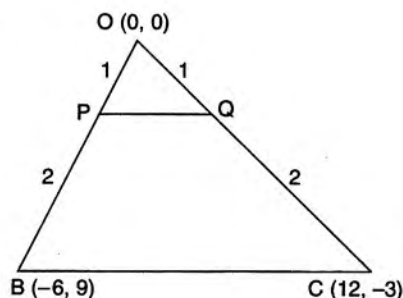
Ans.

- 5** The origin O, B (-6, 9) and C (12, -3) are vertices of triangle OBC. Point P divides OB in the ratio 1 : 2 and point Q divides OC in the ratio 1 : 2. Find the co-ordinates of points P and Q. Also, show that : $PQ = \frac{1}{3} BC$.

Solution :

For point P : $m_1 : m_2 = 1 : 2$, $(x_1, y_1) = (0, 0)$
and $(x_2, y_2) = (-6, 9)$

$$\begin{aligned} \therefore P &= \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) \\ &= \left(\frac{1 \times -6 + 2 \times 0}{1 + 2}, \frac{1 \times 9 + 2 \times 0}{1 + 2} \right) \\ &= (-2, 3) \end{aligned}$$



Ans.

For point Q : $m_1 : m_2 = 1 : 2$, $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (12, -3)$

$$\therefore Q = \left(\frac{1 \times 12 + 2 \times 0}{1 + 2}, \frac{1 \times -3 + 2 \times 0}{1 + 2} \right) = (4, -1)$$

Ans.

Now $PQ =$ Distance between P (-2, 3) and Q (4, -1)

$$= \sqrt{(4 + 2)^2 + (-1 - 3)^2} = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$$

$$\text{and, } BC = \sqrt{(12 + 6)^2 + (-3 - 9)^2} = \sqrt{324 + 144} = \sqrt{468} = 6\sqrt{13}$$

$$PQ = 2\sqrt{13} \text{ and } BC = 6\sqrt{13} \Rightarrow PQ = \frac{1}{3} BC$$

Ans.

- 4** Calculate the ratio in which the line joining the points (4, 6) and (-5, -4) is divided by the line $y = 3$. Also, find the co-ordinates of the point of intersection.

Solution :

The co-ordinates of every point on the line $y = 3$ will be of the type $(x, 3)$.

$$\text{Now, } y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \quad [\text{Taking : } (x, 3) = (x, y), (4, 6) = (x_1, y_1) \text{ and } (-5, -4) = (x_2, y_2)]$$

$$\Rightarrow 3 = \frac{m_1 \times -4 + m_2 \times 6}{m_1 + m_2}$$

$$\Rightarrow 3m_1 + 3m_2 = -4m_1 + 6m_2 \Rightarrow 7m_1 = 3m_2 \Rightarrow \frac{m_1}{m_2} = \frac{3}{7}$$

$\therefore$ The required ratio is 3 : 7

Ans.

$$\text{Now, } x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \Rightarrow x = \frac{3 \times -5 + 7 \times 4}{3 + 7} = \frac{13}{10}$$

$\therefore$ The required point of intersection = $\left(\frac{13}{10}, 3\right)$

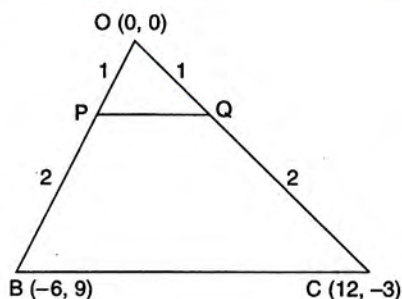
Ans.

- 5** The origin O, B (-6, 9) and C (12, -3) are vertices of triangle OBC. Point P divides OB in the ratio 1 : 2 and point Q divides OC in the ratio 1 : 2. Find the co-ordinates of points P and Q. Also, show that : $PQ = \frac{1}{3} BC$.

Solution :

For point P : $m_1 : m_2 = 1 : 2$, $(x_1, y_1) = (0, 0)$
and $(x_2, y_2) = (-6, 9)$

$$\begin{aligned} \therefore P &= \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) \\ &= \left(\frac{1 \times -6 + 2 \times 0}{1 + 2}, \frac{1 \times 9 + 2 \times 0}{1 + 2} \right) \\ &= (-2, 3) \end{aligned}$$



Ans.

For point Q : $m_1 : m_2 = 1 : 2$, $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (12, -3)$

$$\therefore Q = \left(\frac{1 \times 12 + 2 \times 0}{1 + 2}, \frac{1 \times -3 + 2 \times 0}{1 + 2} \right) = (4, -1)$$

Ans.

Now PQ = Distance between P (-2, 3) and Q (4, -1)

$$= \sqrt{(4 + 2)^2 + (-1 - 3)^2} = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$$

$$\text{and, } BC = \sqrt{(12 + 6)^2 + (-3 - 9)^2} = \sqrt{324 + 144} = \sqrt{468} = 6\sqrt{13}$$

$$PQ = 2\sqrt{13} \text{ and } BC = 6\sqrt{13} \Rightarrow$$

$$PQ = \frac{1}{3} BC$$

Ans.

13.3 Points of Trisection :

Let points P and Q lie on line segment AB and divide it into three equal parts i.e., $AP = PQ = QB$; then P and Q are called **points of trisection** of AB.

- 6** Find the co-ordinates of the points of trisection of the line segment joining the points A (6, -2) and B (-8, 10).



Solution :

Let P and Q be the points of trisection so that $AP = PQ = QB$.

For P :

$$m_1 : m_2 = AP : PB = 1 : 2; (x_1, y_1) = (6, -2) \text{ and } (x_2, y_2) = (-8, 10)$$

$$\therefore x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} = \frac{1 \times -8 + 2 \times 6}{1 + 2} = \frac{4}{3}$$

$$\therefore y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} = \frac{1 \times 10 + 2 \times -2}{1 + 2} = 2$$

$$\therefore \text{Point P} = \left(\frac{4}{3}, 2 \right)$$

Ans.

For Q :

$$m_1 : m_2 = AQ : QB = 2 : 1; (x_1, y_1) = (6, -2) \text{ and } (x_2, y_2) = (-8, 10)$$

$$\therefore Q = \left(\frac{2 \times -8 + 1 \times 6}{2 + 1}, \frac{2 \times 10 + 1 \times -2}{2 + 1} \right) = \left(-\frac{10}{3}, 6 \right)$$

Ans.

- 7** Show that P (3, m - 5) is a point of trisection of the line segment joining the points A (4, -2) and B (1, 4). Hence, find the value of 'm'.

Solution :

P will be a point of trisection of AB if it divides AB in the ratio 1 : 2 or 2 : 1.

$$\text{Since, } x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$\Rightarrow 3 = \frac{m_1 \times 1 + m_2 \times 4}{m_1 + m_2}$$

$$\Rightarrow 3m_1 + 3m_2 = m_1 + 4m_2$$

$$\Rightarrow 2m_1 = m_2 \text{ and } \frac{m_1}{m_2} = \frac{1}{2} \text{ i.e. } m_1 : m_2 = 1 : 2$$

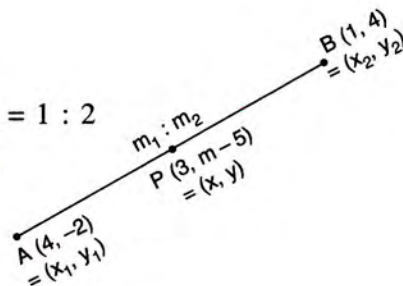
Hence, P is a point of trisection of AB.

$$\text{Now, } y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

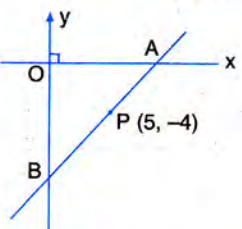
$$\Rightarrow m - 5 = \frac{1 \times 4 + 2 \times -2}{1 + 2}$$

$$\Rightarrow m = 5$$

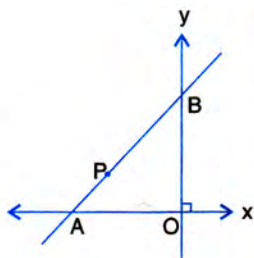
Ans.



EXERCISE 13(A)

- Calculate the co-ordinates of the point P which divides the line segment joining :
 - A (1, 3) and B (5, 9) in the ratio 1 : 2
 - A (-4, 6) and B (3, -5) in the ratio 3 : 2.
- In what ratio is the line joining (2, -3) and (5, 6) divided by the x -axis ?
- In what ratio is the line joining (2, -4) and (-3, 6) divided by the y -axis ?
- In what ratio does the point (1, a) divide the join of (-1, 4) and (4, -1) ?
Also, find the value of a .
- In what ratio does the point (a , 6) divide the join of (-4, 3) and (2, 8) ?
Also, find the value of a .
- In what ratio is the join of (4, 3) and (2, -6) divided by the x -axis ? Also, find the co-ordinates of the point of intersection.
- Find the ratio in which the join of (-4, 7) and (3, 0) is divided by the y -axis. Also, find the co-ordinates of the point of intersection.
- Points A, B, C and D divide the line segment joining the point (5, -10) and the origin in five equal parts. Find the co-ordinates of B and D.
- The line joining the points A (-3, -10) and B (-2, 6) is divided by the point P such that $\frac{PB}{AB} = \frac{1}{5}$. Find the co-ordinates of P.
- P is a point on the line joining A (4, 3) and B (-2, 6) such that $5AP = 2BP$. Find the co-ordinates of P.
- Calculate the ratio in which the line joining the points (-3, -1) and (5, 7) is divided by the line $x = 2$. Also, find the co-ordinates of the point of intersection.
- Calculate the ratio in which the line joining A (6, 5) and B (4, -3) is divided by the line $y = 2$. [2006]
- The point P (5, -4) divides the line segment AB, as shown in the figure, in the ratio 2 : 5. Find the co-ordinates of points A and B.
 
- Find the co-ordinates of the points of trisection of the line joining the points (-3, 0) and (6, 6).
- Show that the line segment joining the points (-5, 8) and (10, -4) is trisected by the co-ordinate axes.
- Show that A (3, -2) is a point of trisection of the line-segment joining the points (2, 1) and (5, -8).
Also, find the co-ordinates of the other point of trisection.
- If A = (-4, 3) and B = (8, -6)
 - Find the length of AB.
 - In what ratio is the line joining A and B, divided by the x -axis ? [2008]
- The line segment joining the points M (5, 7) and N (-3, 2) is intersected by the y -axis at point L. Write down the abscissa of L. Hence, find the ratio in which L divides MN.
Also, find the co-ordinates of L.
- A (2, 5), B (-1, 2) and C (5, 8) are the co-ordinates of the vertices of the triangle ABC. Points P and Q lie on AB and AC respectively, such that : $AP : PB = AQ : QC = 1 : 2$.
 - Calculate the co-ordinates of P and Q.
 - Show that : $PQ = \frac{1}{3}BC$.
- A (-3, 4), B (3, -1) and C (-2, 4) are the vertices of a triangle ABC. Find the length of line segment AP, where point P lies inside BC, such that $BP : PC = 2 : 3$.
- The line segment joining A(2, 3) and B(6, -5) is intercepted by x -axis at the point K. Write down the ordinate of the point K. Hence, find the ratio in which K divides AB.
Also, find the co-ordinates of the point K. [2006]
- The line segment joining A(4, 7) and B(-6, -2) is intercepted by the y -axis at the point K. Write down the abscissa of the point K. Hence, find the ratio in which K divides AB.
Also, find the co-ordinates of the point K.
- The line joining P(-4, 5) and Q(3, 2) intersects the y -axis at point R. PM and QN are perpendiculars from P and Q on the x -axis. Find :
 - the ratio PR : RQ.
 - the co-ordinates of R.
 - the area of the quadrilateral PMNQ. [2004]

24. In the given figure, line APB meets the x -axis at point A and y -axis at point B. P is the point $(-4, 2)$ and $AP : PB = 1 : 2$. Find the co-ordinates of A and B. [2013]

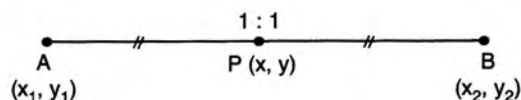


25. Given a line segment AB joining the points $A(-4, 6)$ and $B(8, -3)$. Find :
 (i) the ratio in which AB is divided by the y -axis.
 (ii) find the co-ordinates of the point of intersection.
 (iii) the length of AB. [2012]

13.4 Mid-Point Formula :

To find the co-ordinates of the mid-point of the line segment joining the two given fixed points.

Let P be the mid-point of the line segment joining the points A (x_1, y_1) and B (x_2, y_2) .



Required to find the co-ordinates of P. Suppose $P = (x, y)$.

For mid-point P, the ratio $m_1 : m_2 = 1 : 1$

$$\therefore x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} = \frac{1 \cdot x_2 + 1 \cdot x_1}{1 + 1} = \frac{x_1 + x_2}{2}$$

$$\text{and, } y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} = \frac{1 \cdot y_2 + 1 \cdot y_1}{1 + 1} = \frac{y_1 + y_2}{2}$$

$$\therefore \text{Mid-point of the join of A } (x_1, y_1) \text{ and B } (x_2, y_2) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- 8 Find the co-ordinates of the mid-point of the line segment joining the points P $(4, -6)$ and Q $(-2, 4)$.

Solution :

$$\text{Mid-point} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left(\frac{4 - 2}{2}, \frac{-6 + 4}{2} \right) = (1, -1) \quad \text{Ans.}$$

- 9 The mid-point of line segment AB (shown in the diagram) is $(-3, 5)$. Find the co-ordinates of A and B.

Solution :

Since, point A lies on the x -axis; let $A = (x, 0)$

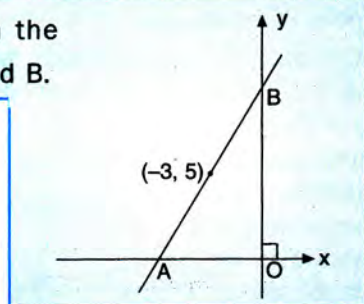
Since, point B lies on the y -axis; let $B = (0, y)$

$$\text{Mid-point of AB} = \left(\frac{x + 0}{2}, \frac{0 + y}{2} \right) = (-3, 5)$$

$$\Rightarrow \frac{x}{2} = -3; \frac{y}{2} = 5 \text{ i.e. } x = -6 \text{ and } y = 10$$

$\therefore$ Co-ordinates of $A = (-6, 0)$ and co-ordinates of $B = (0, 10)$

Ans.



- 10** A (14, -2), B (6, -2) and D (8, 2) are the three vertices of a parallelogram ABCD. Find the co-ordinates of the fourth vertex C.

Solution :

Let C = (x, y)

Since the diagonals of a parallelogram bisect each other;

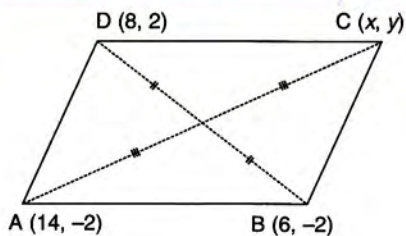
$\therefore$ Mid-point of AC = mid-point of BD

$$\Rightarrow \left(\frac{14+x}{2}, \frac{-2+y}{2} \right) = \left(\frac{8+6}{2}, \frac{2+(-2)}{2} \right)$$

$$\Rightarrow \frac{14+x}{2} = \frac{14}{2} \text{ and } \frac{-2+y}{2} = \frac{0}{2} \Rightarrow x = 0 \text{ and } y = 2$$

$\therefore$ The vertex C = (0, 2)

Ans.



- 11** In triangle ABC, P (-2, 5) is mid-point of AB, Q (2, 4) is mid-point of BC and R (-1, 2) is mid-point of AC. Calculate the co-ordinates of vertices A, B and C.

Solution :

Let A = (x₁, y₁), B = (x₂, y₂) and C = (x₃, y₃).

Since, P is mid-point of AB

$$\Rightarrow \frac{x_1 + x_2}{2} = -2 \text{ and } \frac{y_1 + y_2}{2} = 5$$

$$\text{i.e. } x_1 + x_2 = -4$$

$$\text{and, } y_1 + y_2 = 10$$

Since, Q is mid-point of BC

$$\Rightarrow \frac{x_2 + x_3}{2} = 2 \text{ and } \frac{y_2 + y_3}{2} = 4$$

$$\text{i.e. } x_2 + x_3 = 4$$

$$\text{and, } y_2 + y_3 = 8$$

Since, R is mid-point of AC

$$\Rightarrow \frac{x_1 + x_3}{2} = -1 \text{ and } \frac{y_1 + y_3}{2} = 2$$

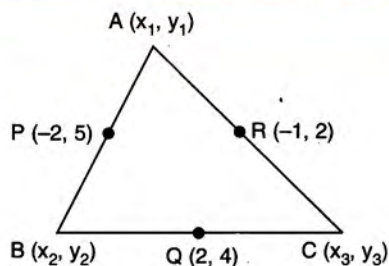
$$\text{i.e. } x_1 + x_3 = -2$$

$$\text{and, } y_1 + y_3 = 4$$

Adding equations I, III and V; we get :

$$x_1 + x_2 + x_2 + x_3 + x_1 + x_3 = -4 + 4 - 2$$

$$\Rightarrow 2(x_1 + x_2 + x_3) = -2$$



.....I

.....II

.....III

.....IV

.....V

.....VI

$$\text{i.e.} \quad x_1 + x_2 + x_3 = -1 \quad \dots \text{VII}$$

On subtracting eq. I from eq. VII, we get : $x_3 = -1 + 4 = 3$

On subtracting eq. III from eq. VII, we get : $x_1 = -1 - 4 = -5$

And, on subtracting eq. V from eq. VII, we get : $x_2 = -1 + 2 = 1$

In the same way, on solving equations II, IV and VI, we get :

$$y_1 = 3, y_2 = 7 \text{ and } y_3 = 1$$

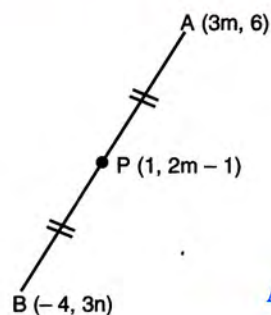
$$\therefore A = (x_1, y_1) = (-5, 3), B = (x_2, y_2) = (1, 7) \text{ and } C = (x_3, y_3) = (3, 1) \quad \text{Ans.}$$

- 12** The mid-point of the line segment joining $(3m, 6)$ and $(-4, 3n)$ is $(1, 2m - 1)$.
Find the values of m and n . [2006]

Solution :

According to the adjoining figure, we have :

$$\begin{aligned} \frac{3m + (-4)}{2} &= 1 \quad \text{and} \quad \frac{6 + 3n}{2} = 2m - 1 \\ \Rightarrow 3m - 4 &= 2 \quad \text{and} \quad 6 + 3n = 4m - 2 \\ \Rightarrow m &= 2 \quad \text{and} \quad 3n = 4m - 8 \\ \Rightarrow 3n &= 4 \times 2 - 8 \\ \Rightarrow m &= 2 \quad \text{and} \quad n = 0 \end{aligned}$$



Ans.

13.5 Centroid of a triangle :

The **centroid** of a triangle is the point of intersection of its medians and it (centroid) divides each median in the ratio 2 : 1.

- 13** Find the co-ordinates of the point of intersection of the medians of triangle ABC; given $A = (-2, 3)$, $B = (6, 7)$ and $C = (4, 1)$.

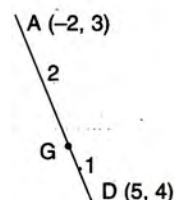
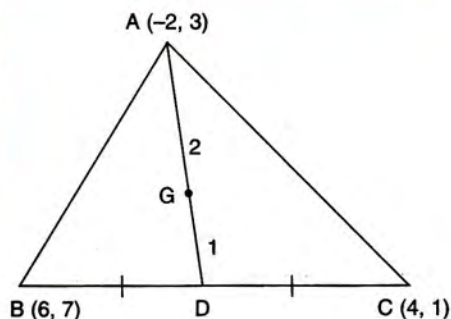
Solution :

Let D be the mid-point of BC.

$$\therefore D = \left(\frac{6 + 4}{2}, \frac{7 + 1}{2} \right) = (5, 4)$$

If G is the point of intersection of medians (centroid), it divides the median AD in the ratio 2 : 1.

$$\begin{aligned} \therefore G &= \left[\frac{2 \times 5 + 1 \times (-2)}{2 + 1}, \frac{2 \times 4 + 1 \times 3}{2 + 1} \right] \\ &= \left(\frac{8}{3}, \frac{11}{3} \right) \end{aligned}$$



Ans.

Direct method : For the vertices A (x_1, y_1), B (x_2, y_2) and C (x_3, y_3) of triangle

$$ABC, \text{ its centroid} = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Thus, in the case of example given above;

$$\begin{aligned} \text{Centroid} &= \left(\frac{-2 + 6 + 4}{3}, \frac{3 + 7 + 1}{3} \right) \\ &= \left(\frac{8}{3}, \frac{11}{3} \right) \end{aligned}$$

Ans.

Taking :

$$(-2, 3) = (x_1, y_1)$$

$$(6, 7) = (x_2, y_2)$$

$$\text{and } (4, 1) = (x_3, y_3)$$

- 14** ABC is a triangle and G(4, 3) is the centroid of the triangle. If A = (1, 3), B = (4, b) and C = (a, 1), find 'a' and 'b'. Find the length of side BC. [2011]

Solution :

Since, G is centroid of ΔABC .

$$\left(\frac{1+4+a}{3}, \frac{3+b+1}{3} \right) = (4, 3)$$

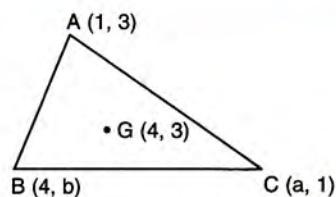
$$\Rightarrow \frac{5+a}{3} = 4 \quad \text{and} \quad \frac{4+b}{3} = 3$$

$$\Rightarrow a = 7 \quad \text{and} \quad b = 5$$

Clearly, B = (4, b) = (4, 5) and C = (a, 1) = (7, 1)

$$\therefore BC = \sqrt{(7-4)^2 + (1-5)^2} = \sqrt{9+16} = \sqrt{25} = 5 \text{ units}$$

Ans.



Ans.

EXERCISE 13(B)

1. Find the mid-point of the line segment joining the points :

(i) (-6, 7) and (3, 5) (ii) (5, -3) and (-1, 7)

2. Points A and B have co-ordinates (3, 5) and (x, y) respectively. The mid-point of AB is (2, 3). Find the values of x and y.

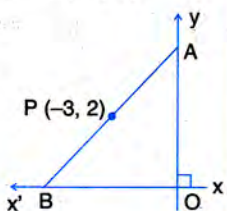
3. A (5, 3), B (-1, 1) and C (7, -3) are the vertices of triangle ABC. If L is the mid-point of AB and M is the mid-point of AC, show that : $LM = \frac{1}{2} BC$.

4. Given M is the mid-point of AB, find the co-ordinates of :

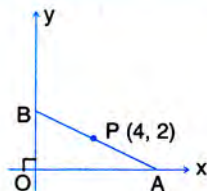
(i) A; if M = (1, 7) and B = (-5, 10),

(ii) B; if A = (3, -1) and M = (-1, 3).

5. P (-3, 2) is the mid-point of line segment AB as shown in the given figure. Find the co-ordinates of points A and B.



6. In the given figure, P (4, 2) is mid-point of line segment AB. Find the co-ordinates of A and B.



7. (-5, 2), (3, -6) and (7, 4) are the vertices of a triangle. Find the length of its median through the vertex (3, -6).

8. Given a line ABCD in which AB = BC = CD, B = (0, 3) and C = (1, 8).

Find the co-ordinates of A and D.

9. One end of the diameter of a circle is (-2, 5). Find the co-ordinates of the other end of it, if the centre of the circle is (2, -1).

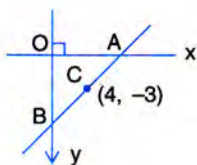
10. A (2, 5), B (1, 0), C (-4, 3) and D (-3, 8) are the vertices of quadrilateral ABCD. Find the co-ordinates of the mid-points of AC and BD. Give a special name to the quadrilateral.

11. P (4, 2) and Q (-1, 5) are the vertices of parallelogram PQRS and (-3, 2) are the co-ordinates of the point of intersection of its diagonals. Find co-ordinates of R and S.

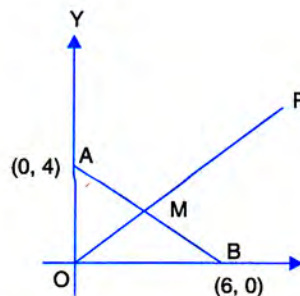
12. A $(-1, 0)$, B $(1, 3)$ and D $(3, 5)$ are the vertices of a parallelogram ABCD. Find the co-ordinates of vertex C.
13. The points $(2, -1)$, $(-1, 4)$ and $(-2, 2)$ are mid-points of the sides of a triangle. Find its vertices.
14. Points A $(-5, x)$, B $(y, 7)$ and C $(1, -3)$ are collinear (*i.e.* lie on the same straight line) such that $AB = BC$. Calculate the values of x and y .
15. Points P $(a, -4)$, Q $(-2, b)$ and R $(0, 2)$ are collinear. If Q lies between P and R, such that $PR = 2QR$, calculate the values of a and b .
16. Calculate the co-ordinates of the centroid of the triangle ABC, if A $= (7, -2)$, B $= (0, 1)$ and C $= (-1, 4)$.
17. The co-ordinates of the centroid of a triangle PQR are $(2, -5)$. If Q $= (-6, 5)$ and R $= (11, 8)$; calculate the co-ordinates of vertex P.
18. A $(5, x)$, B $(-4, 3)$ and C $(y, -2)$ are the vertices of the triangle ABC whose centroid is the origin. Calculate the values of x and y .

EXERCISE 13(C)

1. Given a triangle ABC in which A $= (4, -4)$, B $= (0, 5)$ and C $= (5, 10)$. A point P lies on BC such that $BP : PC = 3 : 2$. Find the length of line segment AP.
2. A $(20, 0)$ and B $(10, -20)$ are two fixed points. Find the co-ordinates of the point P in AB such that $3PB = AB$. Also, find the co-ordinates of some other point Q in AB such that $AB = 6AQ$.
3. A $(-8, 0)$, B $(0, 16)$ and C $(0, 0)$ are the vertices of a triangle ABC. Point P lies on AB and Q lies on AC such that $AP : PB = 3 : 5$ and $AQ : QC = 3 : 5$.
Show that : $PQ = \frac{3}{8} BC$.
4. Find the co-ordinates of points of trisection of the line segment joining the point $(6, -9)$ and the origin.
5. A line segment joining A $(-1, \frac{5}{3})$ and B $(a, 5)$ is divided in the ratio $1 : 3$ at P, the point where the line segment AB intersects the y-axis.
(i) Calculate the value of ' a '.
(ii) Calculate the co-ordinates of 'P'.
6. In what ratio is the line joining A $(0, 3)$ and B $(4, -1)$ divided by the x-axis ?
Write the co-ordinates of the point where AB intersects the x-axis.
7. The mid-point of the segment AB, as shown in diagram, is C $(4, -3)$. Write down the co-ordinates of A and B.
8. AB is a diameter of a circle with centre C $= (-2, 5)$. If A $= (3, -7)$, find
(i) the length of radius AC.
(ii) the coordinates of B. [2013]
9. Find the co-ordinates of the centroid of a triangle ABC whose vertices are :
A $(-1, 3)$, B $(1, -1)$ and C $(5, 1)$. [2006]
10. The mid-point of the line segment joining $(4a, 2b - 3)$ and $(-4, 3b)$ is $(2, -2a)$. Find the values of a and b .
11. The mid-point of the line segment joining $(2a, 4)$ and $(-2, 2b)$ is $(1, 2a + 1)$. Find the values of a and b . [2007]
12. (i) Write down the co-ordinates of the point P that divides the line joining A $(-4, 1)$ and B $(17, 10)$ in the ratio $1 : 2$.
(ii) Calculate the distance OP, where O is the origin.
(iii) In what ratio does the y-axis divide the line AB ?
13. Prove that the points A $(-5, 4)$; B $(-1, -2)$ and C $(5, 2)$ are the vertices of an isosceles right-angled triangle. Find the co-ordinates of D so that ABCD is a square.
14. M is the mid-point of the line segment joining the points A $(-3, 7)$ and B $(9, -1)$. Find the co-ordinates of point M. Further, if R $(2, 2)$ divides the line segment joining M and the origin in the ratio $p : q$, find the ratio $p : q$.
15. Calculate the ratio in which the line joining A $(-4, 2)$ and B $(3, 6)$ is divided by point P $(x, 3)$. Also, find (i) x (ii) length of AP. [2014]



16. Find the ratio in which the line $2x + y = 4$ divides the line segment joining the points $P(2, -2)$ and $Q(3, 7)$.
17. If the abscissa of a point P is 2. Find the ratio in which this point divides the line segment joining the points $(-4, 3)$ and $(6, 3)$. Also, find the co-ordinates of point P .
18. The line joining the points $(2, 1)$ and $(5, -8)$ is trisected at the points P and Q . If point P lies on the line $2x - y + k = 0$, find the value of k . Also, find the co-ordinates of point Q .
19. M is the mid-point of the line segment joining the points $A(0, 4)$ and $B(6, 0)$. M also divides the line segment OP in the ratio $1 : 3$. Find :
 (i) co-ordinates of M
 (ii) co-ordinates of P
 (iii) length of BP



20. Find the image of the point $A(5, -3)$ under reflection in the point $P(-1, 3)$.
21. $A(-4, 2)$, $B(0, 2)$ and $C(-2, -4)$ are vertices of a triangle ABC . P , Q and R are mid-points of sides BC , CA and AB respectively. Show that the centroid of $\triangle PQR$ is the same as the centroid of $\triangle ABC$.